

SANJEEV KUMAR

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EDUCATION

Bachelor of Technology — Computer Science and Engineering — SRM University AP, Andhra Pradesh *CGPA: 8.7 / 10 · Expected 2028*

Class XI – XII, CBSE Board — DAV School, Kathmandu, Nepal

Percentage: 84.32%

Class X — Paribodh Boarding High School, Kathmandu, Nepal

CGPA: 3.20 / 4.00

TECHNICAL SKILLS

Languages	Go (Golang), C, Java, JavaScript (ES6+), SQL, Bash
Frontend	React.js, HTML5, CSS3, Tailwind CSS, Vite, Webpack
Backend	Node.js, Express.js, REST APIs, Microservices Architecture
Databases	MongoDB, SQL
DevOps & Cloud	Docker, Kubernetes, GitHub Actions, CI/CD, Linux
Tools	Git, Postman, Vercel, Netlify, VS Code, Vim

PROJECTS

CommitHub — Custom Version Control System

Go · CLI · File System · Hashing

- Engineering a Git-inspired VCS from scratch in Go: implements init, add, commit, branch, checkout, merge, log, status, and revert
- Designed a content-addressable object store using hashing and tree structures to detect and persist file changes across commits
- Modelled commit history as a Directed Acyclic Graph (DAG) with branch-pointer references for divergence tracking and merge resolution
- Built modular CLI commands with structured error handling and correct edge-case state transitions throughout

Wandera — Travel Experience Platform

React.js, Node.js, Express.js, MongoDB, REST APIs

- Built a full-stack travel platform with modular component architecture; REST APIs handle auth, content management, and location-based search
- Optimised frontend via lazy loading, code-splitting, and dynamic routing; responsive UI implemented with Tailwind CSS

FailureScope (ongoing)

Node.js · JavaScript · DevOps

- Building a log ingestion and failure detection platform that processes application event streams to identify failure patterns in real time
- Applies temporal pattern recognition to correlate event sequences with failure signatures and surface actionable reliability metrics

RESEARCH EXPERIENCE

Nanosensor Network Communication Simulator · MATLAB

- Simulated multi-hop communication architectures for nano-scale sensor networks, modeling node topology, routing, and packet propagation
- Evaluated end-to-end delay, link efficiency, and transmission reliability across variable network densities and hop configurations
- Identified communication bottlenecks under constrained energy budgets — directly applicable to body-area sensor networks (BANs) for wearable monitoring